of spatial information; if we add in time, we get a total of five pieces of spacetime information—one more than we normally would expect.

And so, rather surprisingly, we see that although we are aware of only three extended spatial dimensions, Kaluza's and Klein's reasoning shows that this does not preclude the existence of additional curled-up dimensions, at least if they are very small. The universe may very well have more dimensions than meet the eye.

How small is "small?" Cutting-edge equipment can detect structures as small as a billionth of a billionth of a meter. So long as an extra dimension is curled up to a size less than this tiny distance, it is too small for us to detect. In 1926 Klein combined Kaluza's initial suggestion with some ideas from the emerging field of quantum mechanics. His calculations indicated that the additional circular dimension might be as small as the Planck length, far shorter than experimental accessibility. Since then, physicists have called the possibility of extra tiny space dimensions Kaluza-Klein theory.

2

Comings and Goings on a Garden Hose

The tangible example of the garden hose and the illustration in Figure 8.3 are meant to give you some sense of how it is possible that our universe has extra spatial dimensions. But even for researchers in the field, it is quite difficult to visualize a universe with more than three spatial dimensions. For this reason, physicists often hone their intuition about these extra dimensions by contemplating what life would be like if we lived in an imaginary lower- dimensional universe —following the lead of Edwin Abbott's enchanting 1884 classic popularization Flatland3—in which we slowly realize that the universe has more dimensions than those of which we are directly aware. Let's try this by imagining a two-dimensional universe shaped like our garden hose. Doing so requires that you relinquish an "outsider's" perspective that views the garden hose as an object in our universe. Rather, you must leave the world as we know it and enter a new Garden-hose universe in which the surface of a very long garden hose (you can think of it as being infinitely long) is all there is as far as spatial extent. Imagine that you are a tiny ant living your life on its surface.